11.5 Lines and Planes in Space

To use scalar and vector products to write equations for lines, line segments, and planes in space.

Vector Equation for a Line

Suppose there is a line L through point $P_0(x_0, y_0, z_0)$ and it is parallel to the vector $\mathbf{v}$.

Let P be a point on L, then we know $\overrightarrow{P_0P}$ is parallel to $\mathbf{v}$.

Parametric Equations for a Line

The standard parametrization of the line through $P_0(x_0, y_0, z_0)$ parallel to

$\mathbf{v} = v_1\mathbf{i} + v_2\mathbf{j} + v_3\mathbf{k}$ is

$$x = x_0 + tv_1,$$
$$y = y_0 + tv_2,$$
$$z = z_0 + tv_3,$$

where $t \in \mathbb{R}$.

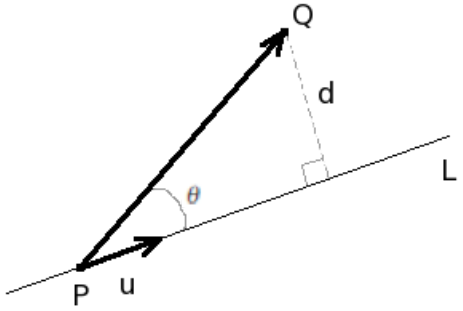
Example: What is the parametric equations for the line through the point $(-2, 0, 4)$ and parallel to the vector

$\mathbf{v} = v_1\mathbf{i} + v_2\mathbf{j} + v_3\mathbf{k}$?

Example: Find the parametric equations for the line through $P(-\ 3, 2, -\ 3)$ and $Q(1, -\ 2, 4)$.

The Distance from a Point to a Line in Space

Suppose L is a line parallel to $\mathbf{u}$ and Q is a point not on L.



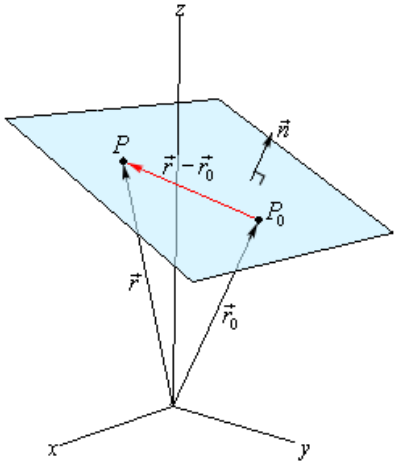
Let P be any point on the line L, then we have

$$d = |\overrightarrow{PQ}| \sin\theta$$

Example: Find the distance from point $S(1, 1, 5)$ to the line

$$L\colon\ x = 1 + t,\ y = 3 - t,\ z = 2t.$$

An Equation for a Plane in Space



A plane in space is determined by a point in the plane and a vector perpendicular or normal to the plane.

Let $P_0(x_0, y_0, z_0)$ be a point in the plane M and let $\mathbf{n} = A\mathbf{i} + B\mathbf{j} + C\mathbf{k}$ be a vector normal to the plane M.

If $P(x, y, z)$ is a point in the plane, then the vector $\overrightarrow{P_0P}$ is perpendicular to $\mathbf{n}$, implying

$$\overrightarrow{P_0P} \cdot \mathbf{n} = 0.$$

Equation for a Plane

The plane passes through $P_0(x_0, y_0, z_0)$ and perpendicular to $\mathbf{n} = A\mathbf{i} + B\mathbf{j} + C\mathbf{k}$ can be represented by

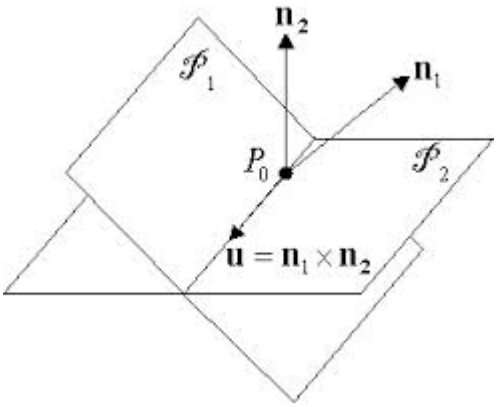
- Vector equation: $\overrightarrow{P_0P} \cdot \mathbf{n} = 0.$
- Component equation: $A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$

Example: The plane passing through $P_0(-3, 0, 7)$ perpendicular to $\mathbf{n} = 5\mathbf{i} + 2\mathbf{j} - \mathbf{k}$ has component equation:

Example:
Find an equation for the plane M through $A(0, 0, 1)$, $B(2, 0, 0)$, and $C(0, 3, 0)$.

Lines of Intersection

Just as lines are parallel if and only if they have the same direction, two planes are *parallel* if and only if their normals are parallel, or $\mathbf{n}_1 = k\mathbf{n}_2$ for some scalar k. Two planes that are not parallel intersect in a line.



Let L be the line of intersection of the planes P_1 and P_2.

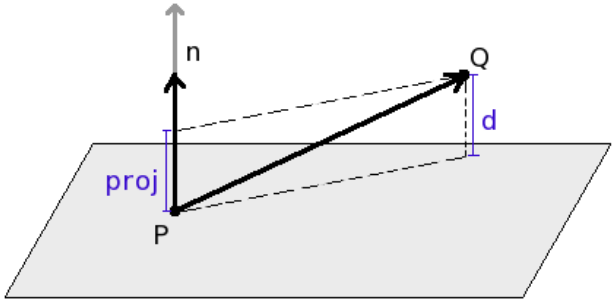
Example: Find the equation of the line of intersection of the planes $3x - 6y - 2z = 15$ and $2x + y - 2z = 5$.

Example: Find the point where the line

$$x = \frac{8}{3} + 2t,\ y = -2t,\ z = 1 + t$$

intersects the plane $3x + 2y + 6z = 6$.

The Distance from a point to a Plane

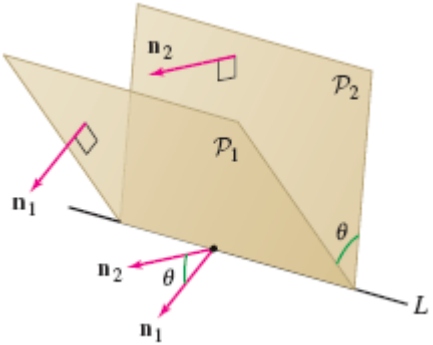


If P is a point (any point) on the plane with normal n, then the distance from any point Q to the plane is the length of the vector projection of $\vec{PQ}$ onto n. That is, the distance from Q to the plane is

$$d = \frac{|\overrightarrow{PQ} \cdot \mathbf{n}|}{|\mathbf{n}|}.$$

Example: Find the distance from $Q(1, 1, 3)$ to the plane $3x + 2y + 6z = 6$.

Angle Between Planes



The angle between two planes can be determined by their normal vectors.

The angle between two planes is the same as the angle between their normal vectors.

Example: Find the angle between the planes $3x - 6y - 2z = 15$ and $2x + y - 2z = 5$.

Example: Let M and N be the planes with plane equations:

$$M: 3x + y + 3z = 12$$

$$N: 2x - 3y + 5z = 19$$

Find the equation of the plane containing the line of intersection of the two planes M and N and the point $(1, 2, 1)$.